

A permanent-polynomial obstruction for monotone comparison tensors

Research note: application of the monotone column permanent theorem

30 September 2026

Let $y_1, \dots, y_m$ be weakly increasing real columns on K rows, with all entries in $[a, b]$. Suppose $K \geq m$ and that for every set $S \subseteq [m]$,

$$\frac{1}{K} \sum_{q=1}^K \prod_{j \in S} y_j(q) = \mu_{|S|}. \quad (1)$$

Write $(u)_s = u(u-1) \cdots (u-s+1)$, and define $c_0, c_1, \dots$ formally by

$$\sum_{s \geq 0} c_s t^s = \exp \left(-K \sum_{j \geq 1} \frac{\mu_j}{j} t^j \right). \quad (2)$$

Only $\mu_1, \dots, \mu_m$ are needed below.

Theorem 1. *The degree- m polynomial*

$$Q_{K,m}(z) = \sum_{s=0}^m \frac{(m)_s}{(K)_s} c_s z^{m-s}$$

has all its roots in $[a, b]$.

Proof. Consider the normalized rectangular permanent

$$R(z) = \frac{1}{(K)_m} \sum_{\iota: [m] \hookrightarrow [K]} \prod_{j=1}^m (z - y_j(\iota(j))).$$

It is real-rooted by the monotone column permanent theorem of Brändén, Haglund, Visontai and Wagner [1, Proposition 4.2]. Indeed, the columns of $(-y_j(q))$ decrease weakly. Equivalently, pad this matrix with $K - m$ zero columns and apply the square theorem: the resulting univariate permanent is a positive constant times $z^{K-m} R(z)$. For $z > b$ every summand defining R is positive, and for $z < a$ every summand has the same nonzero sign. Hence every root of R lies in $[a, b]$.

Fix $S \subseteq [m]$, $|S| = s$. By inclusion-exclusion on equal row indices (the partition-lattice Möbius formula),

$$\sum_{\iota: S \hookrightarrow [K]} \prod_{j \in S} y_j(\iota(j)) = \sum_{\pi \in \Pi(S)} \prod_{B \in \pi} (-1)^{|B|-1} (|B|-1)! K \mu_{|B|}.$$

The exponential formula and (2) show that this quantity is $(-1)^s s! c_s$, independently of the chosen subset S . Each assignment on S extends to $(K-s)_{m-s}$ assignments on all columns. Expanding R therefore gives its coefficient of z^{m-s} as

$$\frac{(-1)^s \binom{m}{s} (K-s)_{m-s}}{(K)_m} (-1)^s s! c_s = \frac{(m)_s}{(K)_s} c_s.$$

Thus $R = Q_{K,m}$. □

Corollary 1. *If $K = m$, the mixed-moment identities (1) imply the existence of K real numbers $t_1, \dots, t_K \in [a, b]$ with*

$$\frac{1}{K} \sum_{q=1}^K t_q^s = \mu_s \quad (1 \leq s \leq K).$$

In particular, a weakly monotone m -column, m -row uniform-interval comparison tensor implies an ordinary real equal-weight degree- m quadrature formula with m nodes.

Proof. Here $Q_{K,K}(z) = \sum_{s=0}^K c_s z^{K-s}$. Its roots belong to $[a, b]$ by Theorem 1. Newton's identities and (2) give the claimed power sums of these roots. $\square$

This statement permits flat columns and repeated row values, and does not assume that many columns coincide. Rational input does not imply rational roots, so it is a real quadrature reduction only.

For centered uniform-interval moments, $[a, b] = [-1, 1]$, $\mu_{2j} = 1/(2j+1)$ and $\mu_{2j-1} = 0$. Thus

$$\sum_{s \geq 0} c_s t^s = \exp \left(-K \sum_{j \geq 1} \frac{t^{2j}}{2j(2j+1)} \right), \quad s c_s = -K \sum_{\substack{2 \leq j \leq s \\ j \text{ even}}} \frac{c_{s-j}}{j+1}.$$

This gives an exact rational algorithm for the obstruction. For any formal degree- K moment polynomial $P_K(z) = \sum_{s=0}^K c_s z^{K-s}$, one also has

$$Q_{K,m}(z) = \frac{m!}{K!} P_K^{(K-m)}(z).$$

In particular differentiation can restore real-rootedness that the full degree- K polynomial lacks.

For example, when $K = m = 8$ the polynomial is

$$z^8 - \frac{4}{3}z^6 + \frac{22}{45}z^4 - \frac{148}{2835}z^2 - \frac{43}{42525}.$$

Its constant coefficient is negative. But an even monic degree-eight polynomial with eight real roots has four pairs of opposite roots, and therefore a nonnegative constant coefficient. This excludes a weakly monotone eight-column, eight-row comparison tensor without any numerical root calculation. Together with the earlier rational bound $K \geq n - 1$, it gives $g(9) \geq 9$.

Exact rational Sturm computations give the following first admissible $K \geq m$ for this necessary condition alone:

m	8	9	10	11	12	14	16	18	20	24	28	30
K	9	9	11	12	14	16	19	22	24	29	35	37.

The complete table through $m = 30$ is produced by `check_monotone_permanent.py`. Passing this polynomial test does not prove that a comparison tensor exists. These data do not support a quadratic lower bound from this obstruction alone.

References

- [1] P. Brändén, J. Haglund, M. Visontai and D. G. Wagner, *Proof of the monotone column permanent conjecture*, Notions of Positivity and the Geometry of Polynomials, Trends in Mathematics, Birkhäuser (2011), 63–78. arXiv:1010.2565, doi:10.1007/978-3-0348-0142-3.5.